14 — Bulk Carrier: L3 Variation from Vessel-Type Lookup

Small bulk carrier from the DB row: curve 12 kn → **1 365 kW**, SM **20 %**, ME 1×12 000 (Diesel 4-stroke Medium), SG 200, AE 2×500. Profile modes: Transit 5 717 h/165 · Port 2 592/110 · Anchor 451/180 · Maneuvering 175 (400/190). No battery. **Load Variation field left EMPTY.**

The point of the test: the L3 lookup

```
HotelLoadVariationKw = empty → backend looks up VesselTypeName:
"Bulk Carrier 10,000 dwt" contains "Bulk Carrier" → appsettings VesselVariations → ±250
DRC reduction 20 %:  250 × 0.8 = 200
IL3 Optimization Details must read:  "Variation: ±250 kW → ±200 kW"   ← the proof
```

(Config map: Bulk Carrier 250 · Container 1500 · LNG 1000 · default 500.)

Power Demands

Transit propulsion = 1 365×1.20 = **1 638**. All hotels via SG 200 (AE = 0 everywhere — the SG-forced rule again; the 12 MW ME idles at ~1 % in port/anchor to spin a 110–200 kW hotel). Energies: Transit (1 638+165)×5 717 = **10 307 751** · Port 110×2 592 = **285 120** · Anchor **81 180** · Maneuvering (400+190)×175 = **103 250**. Total hours 8 935.

Baseline & IL1 — an honest zero

Transit has exactly **one** valid combination (SG 200 covers hotel 165; both 500-kW AEs would sit idle → rejected). One row ⇒ baseline = optimal ⇒ **IL1 = Baseline = 1 883.4 t/yr, savings 0 t / $0** against a $110k investment. The calculator refuses to invent value where a single-engine plant offers no choices. L3 also nets 0 at these load points (SG capacity 200 clamps the spike cycle); the lookup display is the deliverable, not the savings.

Takeaway: VesselTypeName's only direct role in the math is this L3 lookup — and the UI shows it verbatim in the IL3 details line.